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Measurement of Cation Exchange Capacity- A Laboratory Comparative Study

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Abstract

Cation exchange capacity (CEC) plays a critical role in evaluating shaly formations, as it directly influences the contribution of clay minerals and clay-bound water to formation electrical conductivity. There are several laboratory methods currently available to determine CEC and results from these methods are often used interchangeably. This study aims to evaluate and compare the results of those methods, summarize their advantages and limitations, and provide guidelines for laboratory CEC evaluations. Methods of measuring CEC evaluated in this study are multi-salinity Co (core conductivity) and Cw (water conductivity), wet chemistry methylene blue dye test (MBT), and XRD mineralogy based synthetic CEC. The use of a new method based on Inductively Coupled Plasma (ICP) mass spectrometry is also explored.

Based on XRD mineralogy, twelve clastic and four carbonate cores were selected for this study. The Co-Cw method measures the conductivity of samples saturated with brines of varying salinities and CEC is determined by analyzing the relationship between Co and Cw. In the MBT method, the adsorption capacity of solids using a methylene blue dye solution is measured and linked to sample CEC. The XRD mineralogy based synthetic CEC is simply a summation of CECs of individual clay minerals by using a mixing law. The ICP method chemically extracts cations from minerals using ammonium acetate, followed by measuring the concentrations of the extracted Na+, Mg++, K+, and Ca++ ions to determine CEC.

Experimental results show that the CEC values determined by MBT and XRD correlate well, with a correlation coefficient R2 of about 0.85. This is important since XRD mineralogy is commonly acquired, thus abundantly available for reservoir evaluation, so that with XRD clay mineralogy, MBT equivalent CEC can be derived. Differences in CEC values obtained from using the various methods were observed and attributed to sample preparation and the type of samples tested, i.e., core plugs for Co-Cow vs crushed rock powders for MBT, ICP, and XRD methods. Sample preparation is especially important for the methods that use powder samples.

Obviously, CEC from Co-Cw method is more formation representative due to the sample retaining its original distribution of minerals within the rock pore space, but the test needs to be repeated with multiple brine salinities, thus it is time consuming, especially for samples with high clay content; those rock samples with low permeabilities. Proper integration between powder and plug measurements would require

additional data such as petrology to characterize the distribution of minerals. The effects of cations extracted from non-clay minerals on ICP tests are also being explored briefly. This study captures key insights into both the advantages and limitations of the CEC measurement techniques which should be valuable in quality control of CEC data for evaluating shaly formations.

Introduction

The concept of cation exchange and the term of cation exchange capacity (CEC) were coined in the mid-1800s (Keall, 1991), though, its definitions may vary based on measurement techniques. According to Grim (1968), CEC can be defined as the sum of the exchangeable cations that a mineral can adsorb at a specific pH. Chilingarian and Vorabutr (1983) define cation exchange as the cation-for-cation exchange between the clay-building unit and a water solution, which contains the cation of greater attraction. This attraction or replacing power of an ion differs depending upon test conditions, clay type, particle size and valency of the cation (Sanders, et al., 2003). The clay minerals, which carry large number of exchangeable cations are abundant in hydrocarbon-bearing formations, such as shaly sands and organic-rich mud-rocks. The quantity of these positive exchangeable ions per unit weight of dry rock was defined as CEC by Johnson and Linke (1978).

Cations are positively charged elements, the five most abundant exchangeable cations in rocks are calcium (Ca^{++}), magnesium (Mg^{++}), potassium (K^+), sodium (Na^+) and aluminum (Al^{+++}), with the exchanging power in the order of $Na^+ < K^+ < Mg^{++} < Ca^{++} < Al^{+++}$, i.e., K^+ in solution can exchange a Na^+ on clay surface for example. CEC varies depending on the type of clay; with montmorillonite the highest, the heavily weathered kaolinite the lowest, and the less weathered illite in between (Lines-Kelly et al., 1994). As a measure of the ability of a clay to attract and retain cations, if CEC is measured as the number of cations that a clay can exchange with the surrounding solution, in milliequivalents of cations per 100 grams of dry clay, i.e., meq/100g, then CEC values for the commonly encountered subsurface clays are shown below in Table 1. From this table, it is noted that the CEC values of clay minerals are not constant, but vary within a range, depending on geological diagenetic changes, clay formation conditions, variability in crystal structure, particle size and shape, surface roughness (Singer et al., 2023) and porosity, isomorphous substitute, chemical impurities, and layer charge heterogeneity.

Table 1—Range of CEC values of commonly encountered subsurface clay minerals

Clay	Montmorillonite	Illite	Chlorite	Kaolinite
CEC (meq/100g)	80-150	10-40	10-30	5-15
Average CEC_avg (meq/100g)	115	25	20	10

There are several commonly used methods in determining CEC of formation rocks. They are the method of testing on core plug conductivity saturated with brine (Co) vs brine conductivity (Cw), i.e., the Co-Cw method, wet chemistry methylene blue CEC testing (MBT) method, and the inductively coupled plasma (ICP) CEC measurement method. Samples required for both MBT and ICP tests are powder samples, and obviously all those CEC tests are laboratory measurements, though method of extracting CEC from downhole electric logs have also been proposed (Zhang et al., 2018 and 2020).

In clay-bearing rocks, traditional Archie's equation assumes that all electrical conductivity arises from pore water, which leads to overestimation of water saturation if clay effects are not considered. By considering the clay-bound conductivity, the integration of lab-derived CEC yields more representative resistivity corrections and significantly improves accuracy in water saturation calculations. Lab-measured CEC, often converted to the parameter Q_v (CEC per unit pore volume), quantifies the ionic charge exchange potential of clays, enabling models like Waxman–Smits (Waxman and Smits, 1969) and Dual-Water (Dewan and Allix, 1982) to incorporate the additional conductivity introduced by clays. The main objective of this study is to conduct a laboratory comparative study to evaluate the similarities and differences among those commonly used laboratory CEC measurement methods to help users in quality assurance and quality control of CEC data before applications in formation evaluation.

Experiments

In this study, 16 core plugs were selected including sandstone and carbonate with various amount of clays. CT scanning was performed to characterize plug heterogeneity and identify a representative location for subsampling the powder samples for X-ray diffraction (XRD) mineralogy test, as well as the MBT and ICP CEC tests. The XRD data obtained is then used to categorize the samples into four groups in terms of rock type and clay content (Figure 1 and Table 2). The companion core plug sample in each group is saturated with brine and used to conduct the Co-Cw CEC test.

Table 2—Sample grouping based on rock type and clay content

Sample Group	Rock Type	Clay Content (volume or weight %)	ID	Porosity (fraction)	Grain Density (g/cc)	Perm (mD)	CEC Method	
A	Muddy sandstone	10-30%	A01					MBT- Methylene blue test ICP- Inductively coupled plasma XRD- X-ray diffraction
			A02					
			A03					
			PA04	0.21	2.65	671	Co-Cw	
B	Slightly muddy sandstone	5-10%	B01					
			B02					
			PB04	0.29	2.69	808	Co-Cw	
C	Shaly sandstone	30-60%	C01					
			C02					
			C03					
			C04					
			C05					
			C06					
D	Cleaner carbonate	3-7%	D01					
			D02					
			PD04	0.14	2.69	0.18	Co-Cw	

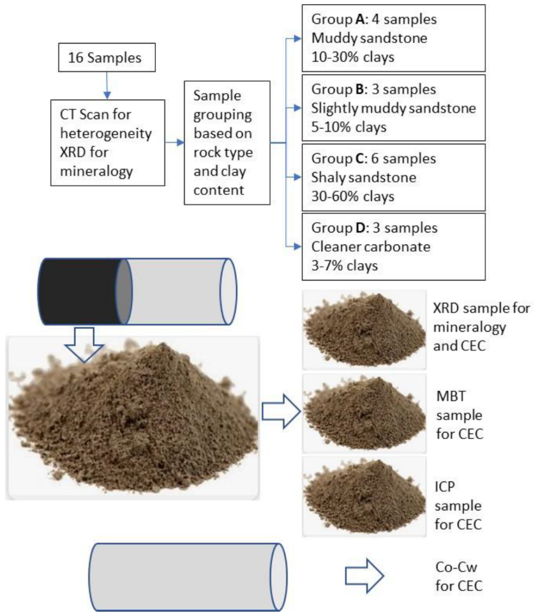

Figure 1—Workflow diagram for sample selection and grouping based on CT scan and mineralogy obtained from XRD analysis.

Methodology

MBT method: MBT is a well-established technique for estimating CEC (Chilingarian and Vorabutr, 1983; Darley and Gray, 1988; API RP 13B-1, 1997; API RP13I; 2000), based on the affinity of the negatively charged clay minerals for positively charged cationic dye. The method is designed for determining the capacity of sample solids to adsorb cations from methylene blue ($C_{16}H_{18}ClN_3S$, MB+ with a molar mass of about 319.85 g/mol) dye solution, which contains a monovalent cation dye that each mole of MB dye corresponds to 1 mole of exchangeable sites, i.e., 1 equivalent.

In conducting the test, the dye is titrated against a dispersed solution of sample solids to be tested. The mixture is constantly stirred, and drops of the suspension are periodically placed on filter paper. The end point is reached when a blue halo ring is formed around the drop sampled out of the suspension, indicating that all exchange sites are saturated and excess dye is present in solution. This ring is the free, un-adsorbed dye and represents the endpoint of the titration, i.e. the total exchange capacity of the sample (Figure 2). The volume of each drop is 0.5 ml, thus the MBT is a semi-quantitative method.

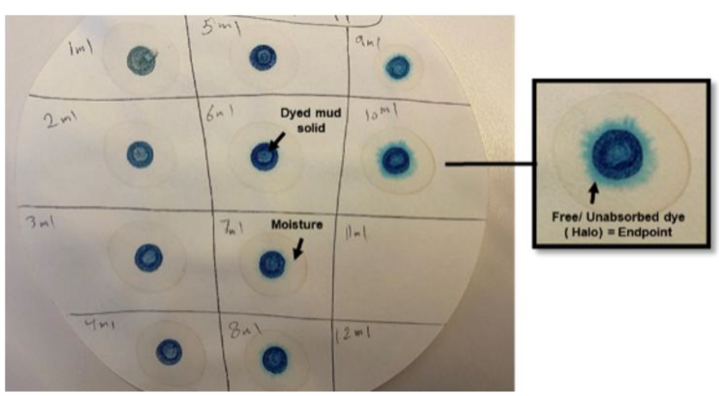

Figure 2—Example Methylene blue dye endpoint in sample titration.

ICP method: Measurements of CEC using the ICP method require quantification of capacities for each exchangeable cation, i.e., the concentrations of Na+, K+, Mg2+, and Ca2+ in water-based solutions.

Ammonium acetate was used to chemically extract cations from clay minerals (Chapman, 1965). The concentrations of Na^+, Mg^{++}, K^+, and $Ca^{++}$are then measured using an ICP mass spectrometry (ICP-MS) to derive CEC of clay minerals (Cheng and Heidari, 2017), thus the process of data acquisition, processing, and interpretation is much more complicated than the MBT method. The 16 samples prepared for the ICP test is shown in Figure 3.

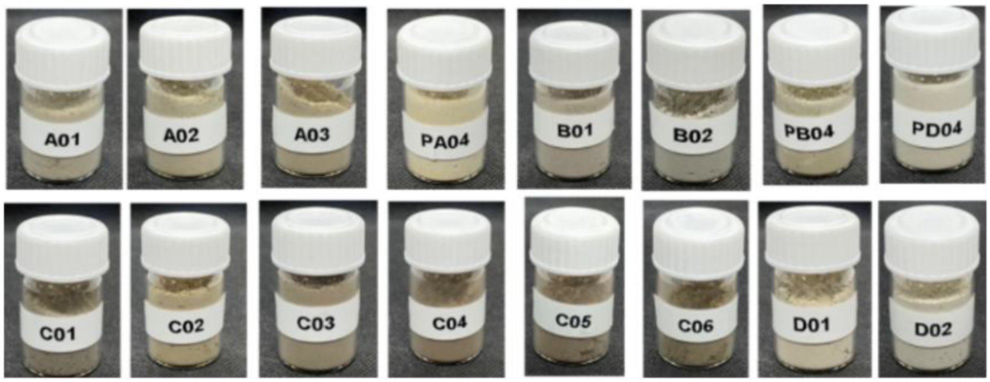

Figure 3—Prepared powder rock samples to be digested for ICP-MS analysis.

Co-Cw method: To characterize CEC of a core sample, parameter Q_v is introduced which is a rock property to quantify the excess conductivity contribution of pore-lining clays in rocks (Hill and Millburn, 1956), i.e., the clay conductivity. Estimating CEC from Q_v involves measuring the conductivity of samples 100% saturated with brines of different salinities, which requires completely cleaning and drying oil reservoir rocks for testing. In this study, three core plugs were selected, preparation for a Co-Cw test requires performing a thorough cleaning of the testing plugs, especially if the rock samples are from oil reservoirs, with the preference of flow through method (Ma et al., 2013; Singer et al., 2022; Qatari et al., 2024). Following the sample preparation, routine core analysis is conducted for porosity, grain density, and permeability, before the Co-Cw test that involves measuring Q_v. With data from routine core analysis (porosity and grain density) and Co-Cw test, CEC can be calculated (Figures 4 and 5).

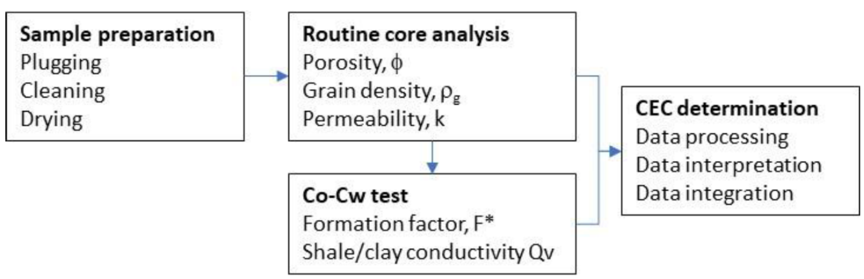

Figure 4—Sample preparation and workflow followed for the Co-Cw method.

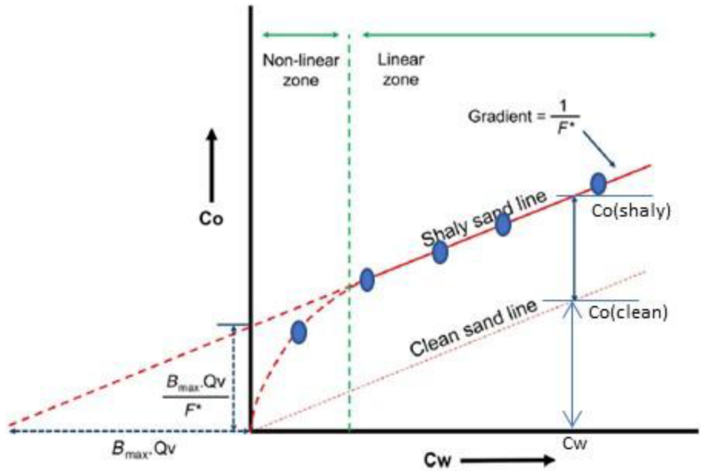

Figure 5—Co-Cw plot for Qv determination.

Once a Co-Cw test is conducted, CEC can be calculated by plotting Co vs Cw and linearly extrapolating the experimental data to Co=0, intercepts of the linear regression line of shaly sand sample data to the x-axis and y-axis are

$$-I_x = B_{max}Q_v \tag{1}$$

and

$$I_y = \frac{B_{max}Q_v}{F^*} = \frac{-I_x}{F^*} \tag{2}$$

Where F* is the formation factor of shaly sand rocks.

Obviously, for clean rock samples, Q_v=0, then both the intercepts are also zero, i.e., the extrapolating line goes through the origin point of the Co-Cw plot. With known water resistivity R_w and clean rock Archie formation factor F

$$F = \frac{R_o}{R_w} = \frac{C_w}{C_o} \tag{3}$$

Formation factor of a shaly sand rock can be obtained

$$F^* = F\left(1 + R_w Q_v B_{max}\right) = F\left[1 + R_w\left(-I_x\right)\right] \tag{4}$$

Once F* is determined, Q_v, the effective concentration of clay-exchange cations per unit pore volume (Dacy and Martin, 2008), may be calculated as

$$Q_v = \frac{B_{max}Q_v}{F^*}\frac{F^*}{B_{max}} = I_y\frac{F^*}{B_{max}} = I_y\frac{F\left[1 + R_w\left(-I_x\right)\right]}{B_{max}} \tag{5}$$

Therefore, Qv can be calculated directly by dividing the value of y-intercept by B_{max}:

$$Q_v = \frac{B_{max}Q_v}{B_{max}} \tag{6}$$

Where B_{max} is a constant depending on temperature T and water resistivity R_w

$$B_{max} = \frac{-1.28 + 0.225\mathrm{T} - 0.0004059\mathrm{T}^2}{1 + (0.045\mathrm{T} - 0.27)R_w^{1.23}} \tag{7}$$

With known Q_v, CEC can be calculated based on

$$\mathrm{CEC} = \frac{100 Q_v \phi}{\rho_g\left(1 - \phi\right)} \tag{8}$$

Where ρ_g is grain density.

Procedures

Methylene Blue Test (MBT) for CEC

The following steps are followed to conduct the MBT CEC tests:

1. **Plug sample cleaning and drying**. Samples were cleaned and dried in an oven at 220 °F.
2. **Powder sample grinding**. The samples were crushed and finely ground to pass through a 75 μm sieve.
3. **Mixture suspension preparation**. Transfer 1g of finely ground powder sample into a test flask. Add 15 ml of Hydrogen Peroxide solution and add 1 ml of 5N Sulphuric acid to the powder sample in the flask. The sample and the solution were well mixed. The flask was then placed on a hot plate, brought the solution to boil and boiled gently for a further 10 minutes.
4. **Sample dilution with water**. Flask was removed from the hot plate, wait until temperature slightly cooled down, and diluted the mixture to 50 ml with distilled water.
5. **Titration for CEC**. Add MB drop (0.5 ml) to the mixture sample and agitate the contents for approximately 20 seconds. CEC titration started by transferring a drop of the solution mixture on to the filter paper with the stirring rod. This process was repeated with each addition of the 0.5 ml MB drop titration, until a blue halo of unabsorbed dye appears at the edge of the drop. Record the number of drops of the titration solution used.
6. **Results for CEC estimation**. The volume of absorbed MB solution was calculated from the number of drops of the titration solution recorded and used to estimate the testing sample CEC (Figure 6).

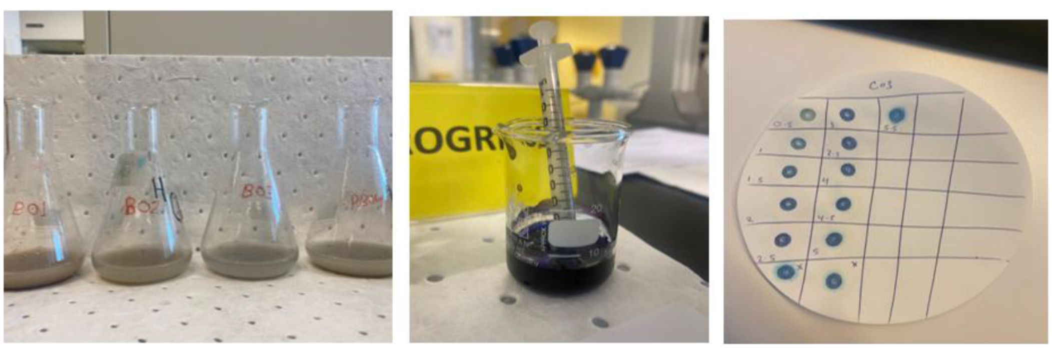

Figure 6—Photos of MBT test procedure performed on prepared rock powder sample and mixed with solution for titration to calculate CEC.

Inductively Coupled Plasma (ICP) Test for CEC

The following steps are followed to conduct the ICP CEC tests:

1. **Plug sample cleaning and drying**. Samples were cleaned and dried in an oven at 220°F.
2. **Powder sample grinding**. The samples were crushed and finely ground to pass through a 75 μm sieve.
3. **Sample Digestion**. Transfer 2g of finely ground powder sample into a test beaker (100 ml). Ammonium acetate of 25 mL with concentration of 1 mol/L is added to the beaker to leach cations from the testing solid sample. The suspension is stirred and kept overnight. Buchner funnel and vacuum pump were used to filter the suspension.
4. **Sample leaching**. The filtered solid sample is leached 3 more times with 25 mL increments of ammonium acetate, with a total of 100 ml ammonium acetate solution collected.
5. **Leachate dilution with water**. The leachate is transferred to a 250 mL volumetric flask and diluted to 250 mL by adding 150 ml deionized water.

6. **Leachate dilution with 2% nitric acid solution**. The leachate is diluted 1000 times by 2% nitric acid solution for ICP measurements (Figure 7).
7. **ICP test for CEC**. The prepared solutions were run along with calibration solutions of 1000 ppm Na^+, Mg^{++}, K^+, and Ca^{++} from Inorganic Ventures into the ICP.

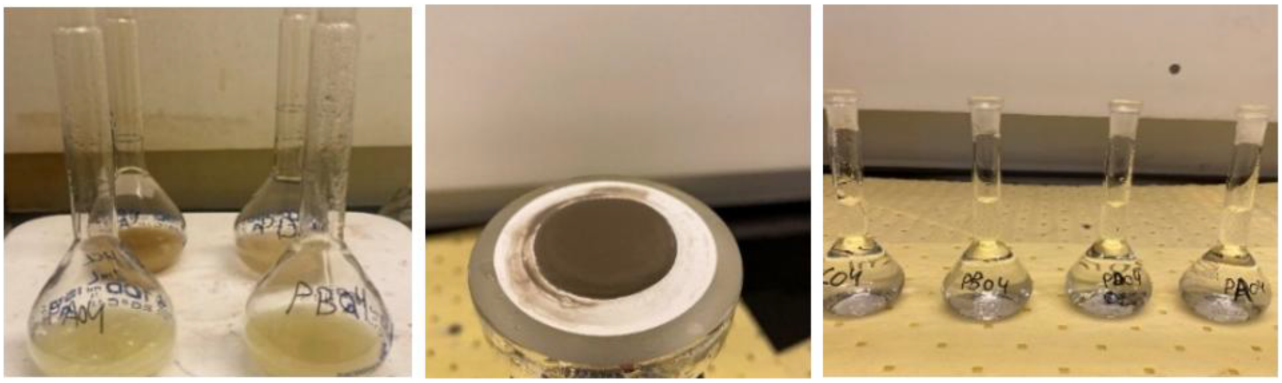

Figure 7—Photos of sample preparation during digestion step for ICP analysis method using ammonium acetate solution.

Multiple Salinity Co-Cw Method for CEC Measurements

For the Co-Cw test core samples, the plugs were cold cleaned (at 40°C) using flow through cleaning to avoid destruction of clay minerals. Toluene was used first then followed by methanol. After cleaning, porosity, permeability, and grain density were measured. The Co-Cw tests were conducted using sodium chloride (NaCl) brine at three differing salinities of 200k ppm, 140k ppm, and 90k ppm. The samples were saturated with the most saline brine first, and then individually mounted into 2-electrode high-pressure resistivity cells (Ma et al., 2013). Using oil as the confining fluid, a net confining stress of 3000 psi was applied. The saturating brine was introduced into the brine saturated core samples with 300 psi backpressure, displacing any remaining trapped gas in the system. Core sample conductivities (Co) were determined until constant values were achieved (+/-1%) with phase angles at <2 degrees. The core samples were shut in while the measurements were taken. The core samples were saturated with the next brine in order of decreasing salinity. Data generated from the above were plotted as Co vs Cw, and the slope of the linear regression line defines 1/F* with the absolute value of intercept on the x-axis is defined as the term $BmaxQ_v$.

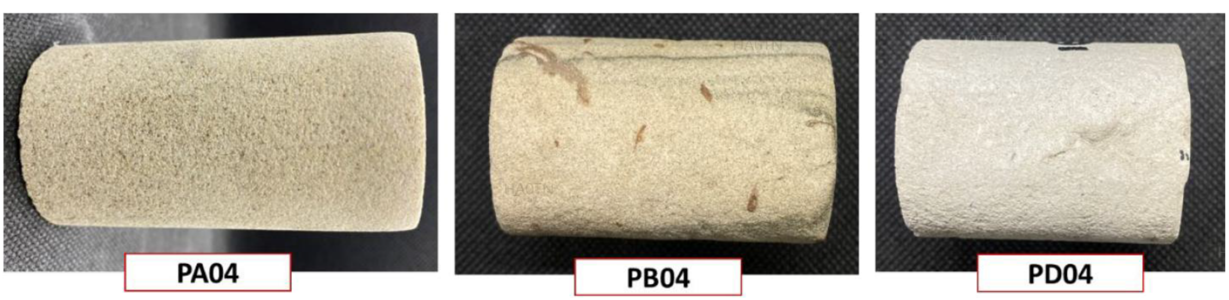

Figure 8—Selected plug samples for the Co-Cw measurements with brine salinities of 200k, 140k, and 90k ppm.

Figure 9—Flow through core cleaning unit (left) and conventional core analysis system for porosity and permeability measurements (right).

Experimental Results

Results of MBT Method

The total amount of MB+ cation adsorbed onto the solid sample is equal to the sum of the amounts adsorbed by each of the clay minerals. It is therefore an additive process representing the clay types (Kariuki et al., 2003), thus CEC can be calculated as

$$\mathrm{CEC} = \frac{\text{MB solution reading (ml)}}{\text{Weight of test solid sample (g)}} \tag{9}$$

In this study, the test was done at every 0.5 ml drop. To test repeatability, measurements were repeated 3 times. The repeated results were selected at least twice as final CEC values (Table 3).

Table 3—MBT tests were repeated 3 times for CEC as per 1 gm of solid sample

Sample ID	Rock Type	Clay content	CEC Test (meq/100g)	CEC Test Repeat 1 (meq/100g)	CEC Test Repeat 2 (meq/100g)	CEC_MBT (meq/100g)
A01	Muddy sandstone	10-30%	4.0	3.5	3.5	**3.5**
A02			2.5	2.5	3.0	**2.5**
A03			4.0	4.0	4.0	**4.0**
PA04			1.0	1.0	0.5	**1.0**
B01	Slightly muddy sandstone	5-10%	1.5	1.5	1.5	**1.5**
B02			2.0	2.0	2.0	**2.0**
PB04			1.0	1.0	0.5	**1.0**
C01	Shaly sandstone	30-60%	5.0	6.0	6.0	**6.0**
C02			5.0	5.0	5.0	**5.0**
C03			5.0	5.0	5.5	**5.0**
C04			2.5	3.0	3.0	**3.0**
C05			3.0	3.0	3.0	**3.0**
C06			9.0	8.5	8.5	**8.5**
D01	Cleaner carbonate	3-7%	1.0	1.0	1.0	**1.0**
D02			1.5	1.0	1.5	**1.5**
PD04			1.5	2.0	2.0	**2.0**

Results of ICP Method

With the ICP measurements, the concentration of each of the cations Na^+, Mg^{++}, K^+, and Ca^{++}values in PPM are obtained and used for sample CEC calculations, after unit conversion from ppm to percentage (Table 4).

$$CEC_{Na^+} = \frac{V_{leach} . PPM_{Na^+}}{Wt_{rock} .\ m_{Na^+}} \tag{10}$$

$$CEC_{Mg^{++}} = 2\frac{V_{\text{leach}} . PPM_{Mg^{++}}}{Wt_{rock} .\ m_{Mg^{++}}} \tag{11}$$

$$CEC_{K^+} = \frac{V_{leach} . PPM_{K^+}}{Wt_{rock} . m_{K^+}} \tag{12}$$

$$CEC_{C^{++}} = 2\frac{V_{leach} . PPM_{Ca^{++}}}{Wt_{rock} . m_{Ca^{++}}} \tag{13}$$

Where Wt_{rock} is the sample weight (i.e., 2g in this study), V is the volume of leachate (i.e., 250 ml in this study), and m is the atomic mass in g/mol of Na^+, Mg^{++}, K^+, and Ca^{++}.

Table 4—CEC values by the ICP method

Sample ID	Rock Type	Clay content	ICP Na, ppm	ICP Mg, ppm	ICP K, ppm	ICP Ca, ppm	CEC_ ICP (meq/100g)
A01	Muddy sandstone	10-30%	4,155	354	279	7,685	7.5
A02			3,788	318	210	2,917	4.3
A03			3,703	451	310	3,752	4.9
PA04			73	81	88	614	0.5
B01	Slightly muddy sandstone	5-10%	561	119	199	728	0.9
B02			2,399	932	338	7,752	7.2
PB04			565	222	90	580	0.9
C01	Shaly sandstone	30-60%	2,912	438	357	3,553	4.4
C02			3,435	420	282	3,387	4.5
C03			2,259	340	410	1,018	2.3
C04			2,514	441	314	2,460	3.5
C05			1,945	1,104	227	2,347	3.7
C06			4,699	623	711	2,076	4.7
D01	Cleaner carbonate	3-7%	1,052	593	146	24,888	16.8
D02			1,596	306	185	28,369	18.9
PD04			1,045	346	125	25,436	16.8

The total sample CEC is the sum of CEC for each exchangeable cation in the sample (Cheng et al., 2018), as listed in Table 5.

$$CEC = CEC_{Na^+} + CEC_{Mg^{++}} + CEC_{K^+} + CEC_{Ca^{++}} \tag{14}$$

Table 5—CEC values calculated by the Co-Cw method

Sample ID	Brine salinity, kppm	200 Kppm	140 Kppm	90 Kppm	Ix, S/m	Iy, S/m	F*	Qv	ϕ (fraction)	k, mD	$\rho_g(1-\phi)/\phi$	CEC (meq/100g)
	Bmax,ml/meq.ohm-m											
	Rw@80C, ohm-m	0.021	0.026	0.037								
	Cw@80C, S/m	47.71	37.96	27.05								
PA04	Co@80C, S/m	2.69	2.13	1.58	-2.19	0.12	18.6	0.15	0.21	671	9.9691	1.61
	F/Cw/Co=	*17.74*	*17.82*	*17.12*								
PB04	Co@80C, S/m	4.08	3.29	2.39	-2.60	0.21	12.2	0.18	0.29	808	6.5859	2.83
	F/Cw/Co=	*11.69*	*11.54*	*11.32*								
PC04	Co@80C, S/m	1.00	0.80	0.57	-0.33	0.01	48.1	0.02	0.14	0.18	16.5243	0.15
	F/Cw/Co=	*47.71*	*47.45*	*47.46*								

Results of Co-Cw Method

In conducting a Co-Cw test, core plug is first cleaned, dried and completed saturated with a specific brine of conductivity Cw. The sample conductivity (Co) was then measured and plotted against Cw. Variation of Co with Cw is usually linear. To determine the linear relationship, a minimum of three data points with three brine salinities is required. Figure 10 and Table 5 show Co-Cw plots for the three plugs tested in this study. Note that because the test requires cleaning, drying, and saturation, the test is not practical for tight rocks, such as the Group C very shaly sandstone samples that permeability is low (Table 2).

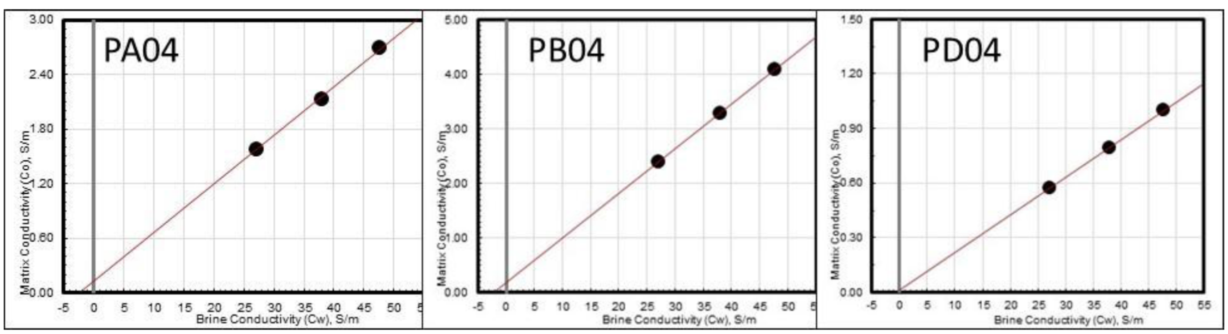

Figure 10—Co-Cw plot for plugs PA04, PB04, and PD04 at 80 °C and 1500 psi

With Q_v determined from the Co-Cw test, sample CEC can be calculated by using Eq. 8 with porosity and grain density measured from routine core analysis. It should be noted that the grain density for the carbonate sample is slightly lower than expected, a phenomenon has been observed often (Ma et al., 2002 and 2013). A detailed study investigating grain density measurement of carbonate rocks has been conducted (Al-Hamad, et al., 2025).

Equations 3, 4, 6, 7, and 8 were used with the obtained values from the Co-Cw plots in Figure 10 to calculate the CEC in Table 5.

CEC Results of XRD Mineralogy

XRD is routinely used to characterize rock mineralogy. To obtain quality mineralogy data, careful planning and calibration are required (Al-Ofi and Ma, 2023). Table 6 listed the XRD results for the 16 samples. Once clay mineralogy is known, and with given CEC for each clay mineral, sample CEC may be calculated using the simple clay volume based mixing rule (Table 7).

$$CEC = \sum_{i=1}^{n} v_i \, CEC_i \qquad (15)$$

Where n is the Number of components, i is the index representing each component (1 to n), v_i is the volume or mass fraction of component I, and CEC_i is the cation exchange capacity of component i. designation.

Table 6—XRD mineralogy (%)

Sample#	Quartz	Calcite	Dolomite	Pyrite	Halite	Siderite	k-feldspares	Albite	Plagioclase	Total clays	Rock Type	Smectite	Illite	Kaolinite	Chlorite
A01	63	0	1	4	0	3	5	0	0	24	Muddy sandstone	4	2	12	6
A02	68	0	0	0	1	0	9	0	0	22		3	6	2	11
A03	70	0	1	0	1	0	7	0	0	21		4	4	9	4
PA04	74	0	0	0	0	0	7	0	5	14		1	5	7	1
B01	89	1	0	0	0	0	0	0	0	10	Slightly muddy sandstone	4	2	3	1
B02	76	3	3	1	0	0	8	0	0	9		1	3	3	2
PB04	91	0	0	0	1	3	0	0	0	5		1	3	1	1
C01	40	0	0	2	0	0	0	0	0	58	Shaly sandstone	5	4	46	3
C02	56	0	0	0	0	2	8	0	0	34		8	8	14	4
C03	48	0	0	0	0	3	0	3	0	46		6	20	12	8
C04	50	0	0	4	0	2	0	0	0	43		3	5	34	1
C05	47	0	8	4	0	3	2	0	0	36		3	3	26	4
C06	39	0	6	1	0	3	2	0	0	49		9	7	27	6
D01	6	70	16	4	0	0	0	0	0	4	Cleaner carbonate	2	2	0	0
D02	1	91	0	1	0	0	0	0	0	7		2	4	1	0
PD04	0	93	0	0	0	0	0	0	0	6		2	2	2	0

Table 7—CEC values from XRD mineralogy, MBT, ICP, and Co-Cw methods

ID	Rock Type	Clay content	Smectite, %	Illite, %	Kaolinite, %	Chlorite, %	CEC_avg	CEC_min	CEC_MBT	CEC_ICP	CEC_CoCw
A01	Muddy sandstone	10-30%	4	2	12	6	8.1	4.6	3.5	7.5	
A02			3	6	2	11	6.5	4.2	2.5	4.3	
A03			4	4	9	4	7.8	4.5	4.0	4.9	
PA04			1	5	7	1	3.9	1.7	1.0	0.5	1.61
B01	Slightly muddy sandstone	5-10%	4	2	3	1	5.8	3.7	1.5	1.0	
B02			1	3	3	2	2.7	1.5	2.0	7.2	
PB04			1	3	1	1	2.0	1.2	1.0	0.9	2.83
C01	Shaly sandstone	30-60%	5	4	46	3	16.3	7.0	6.0	4.4	
C02			8	8	14	4	14.4	8.3	5.0	4.5	
C03			6	20	12	8	15.1	8.2	5.0	2.3	
C04			3	5	34	1	11.6	4.7	3.0	3.5	
C05			3	3	26	4	9.8	4.4	3.0	3.7	
C06			9	7	27	6	18.1	9.9	8.5	4.7	
D01	Cleaner carbonate	3-7%	2	2	0	0	2.8	1.8	1.0	16.8	
D02			2	4	1	0	3.5	2.1	1.5	18.9	
PD04			2	2	2	0	3.2	1.9	2.0	16.8	0.15

As stated in Table 1, CEC for clay minerals is not a constant, but a range. Table 7 documented that sample CEC calculated by assuming the lower limit and average for each clay mineral, together with CEC values derived from the MBT and ICP methods, as well as the three data points for the three plug Co-Cw CEC values. CEC values determined by using the above four different methods, MBT, ICP, Co-Cw, and XRD as shown in Table 6, are plotted in Figure 11. Advantages and limitations of the four CEC measurement methods are summarized in Table 8.

Table 8—CEC measurement methods comparison

	MBT	XRD	CO-CW	ICP
Principle	Based on methylene blue ($C_{16}H_{18}ClN_3S$, MB+) cation exchange with cations associated with clay minerals. The molecule has a flat planar shape with a size of about 2 nm. The cation strongly adsorbs clay minerals and exchanges with other cations.	Based on mineralogy characterization and CEC of each individual clay minerals.	Based on the presence of clay minerals and their contribution to increasing electrical conductivity.	Based on cation exchange between ammonium ion (NH_4^+) of ammonium acetate (CH3COONH4) and the cations The molecule has a spherical shape with a size of about 0.4 nm. The cation weakly adsorbs clay minerals and easily (due to its smaller size and high mobility) exchange with other cations.
Assumptions	Adsorption is monolayer. Clay surface area remains constant during the exchange process MB+ only adsorbs on to clay surface, not penetrate clay structure.	Assuming CEC of each individual clay minerals.	Extra electric conductance is completely due to clay minerals.	The plasma is stable and consistent during the test. The ICP instrument response is linearly related to the concentration of cations in the solution.
Advantages	Quick Simpler lab set up Cost effective.	Data is widely available Calculation is straight forward.	Plug samples are more representative to reservoir rock than crushed powder samples, in terms of clay distributions in the pore space.	High detection limits for many elements Ability to measure more than element simultaneously.
Limitations	Sample preparation (milling and sieving) can affect the quality of test as clay particles are captured at curtain size Results are based on 1- 2 gm of sample which can be missing the representative clays.	Accuracy of mineral identification and quantification Representativeness of CEC of each clay minerals.	Difficulties conducting the test, especially for those with high clay content that permeability is usually low or very low. Presence of electric conductive non-clay minerals will distort the CEC results.	Sample preparation can affect the quality of test. Results are based on 1- 2 gm of sample which may not be representative Requires ICP machine which can be costly NH4+ may be over exchange resulting in higher than expected CEC.

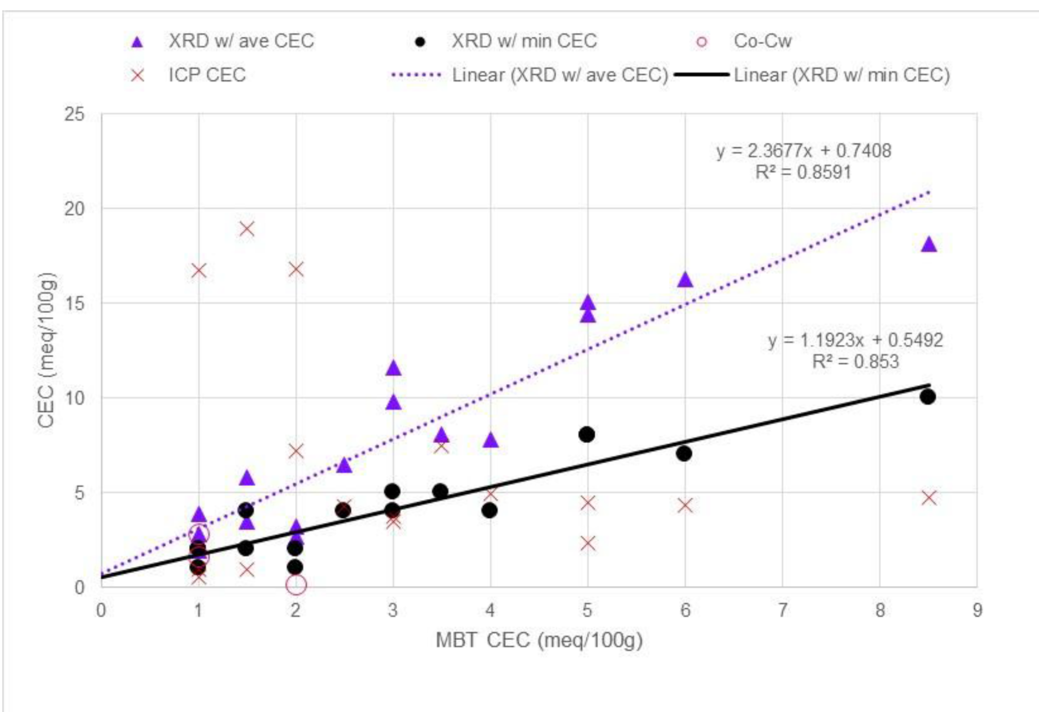

Figure 11—CEC values determined by MBT, ICP, CO-CW, and XRD methods.

Observations and Discussions

Relationship Between CEC_MBT and CEC_XRD

From the summary Figure 11, it is interesting to observe the following:

1. Correlations between the CEC values from the routinely conducted wet chemistry based MBT method and that calculated from the XRD mineralogy are good, either by assuming the average CEC values for individual minerals (triangle, with a correlation coefficient R2=0.8591) or even the min limit of mineral CEC values (dots, R2=0.853).

$$CEC_{XRD-min} = 1.1923\, CEC_{MBT} + 0.5492 \tag{16}$$

and

$$CEC_{XRD-ave} = 2.3677\, CEC_{MBT} + 0.7408 \tag{17}$$

With the above relationships established, MBT equivalent CEC can be estimated from mineralogy measurements such as XRD, for data of mineralogy is much more widely available than wet chemistry CEC values.

2. In principle, the Co-Cw measurements are more representative because it does not require disaggregating the sample mechanically that destroys rock pore structure and mineral distribution where pore fluids reside and interact. In practice, however, saturating and completely displacement of pore brine by another is a time-consuming process, especially for low permeability samples. In addition, change in Co may not be completely due to the presence of clays, if electric conductive minerals such as pyrite/siderate exist in the rock sample, which can be detected from mineralogy tests such as XRD (Table 6) or viewed from images of thin sections and/or SEM. In any case, Co-Cw CEC values are not routinely acquired, and cannot be acquired for extremely low permeability samples, such as those with high clay content such as the Group C samples (Table 6).
3. With the above observations, it seems rational to combine the good correlations between CEC values from MBT and XRD (Eqs. 16 and 17) and calibrate with the representative plug Co-Cw CEC measurement to provide a practical solution to allow users to generate abundant reservoir rock representative CEC values from the widely available mineralogy data, helping to address challenges in shaly reservoir evaluation.
4. The above proposed approach is likely requiring local calibration depending on clay mineral distribution in the pore space and rock matrix.

CEC From ICP Method

As for the ICP method, data is very scattered as shown in Figure 11, mainly due to the way the cations are extracted or leached by using the method. Leaching can not only extract cations from clay minerals, but also from other minerals such as calcite (Ahmee et al., 2020). Apparently, the ICP method is not recommended for carbonate rocks, for the existence of Ca++ in the rock matrix and the tendency of over exchanging with cations in the solution, as illustrated by Figure 12. Interference from other minerals such as pyrite and k-feldspar can also affect the results of CEC from the ICP method.

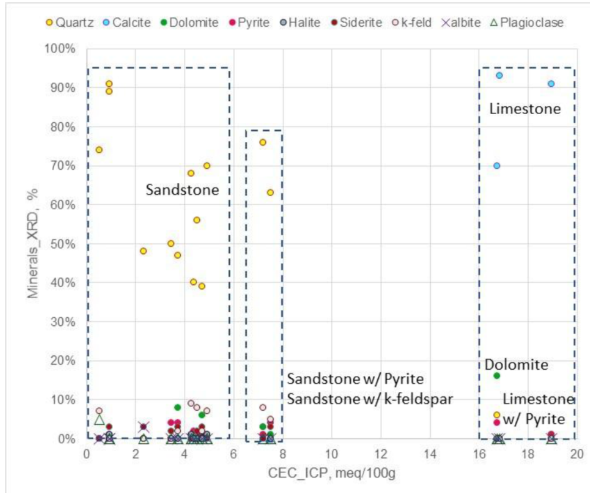

Figure 12—Effect of mineralogy on CEC measured by the ICP method

Focusing on sandstone samples, a trend between CEC-ICP and CEC-MBT may be established, Figure 13, though the quality of correlation is not robust, with the limited data available in this study.

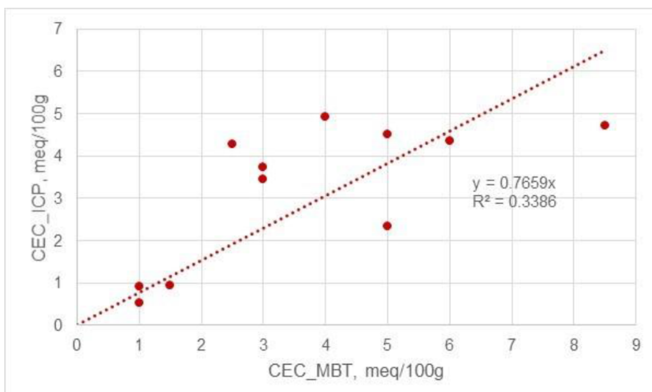

Figure 13—Correlation between CEC_MBT and CEC_ICP

Conclusions

Based on this comparative study of determining rock sample CEC using different methods, the following are concluded:

Good correlations of CEC values derived from MBT and XRD methods are obtained. This could be significant since XRD is widely available, so that with the established correlations, XRD based MBT equivalent CEC can help to improve shaly reservoir evaluation, especially after local calibration with plug Co-Cw results.

Plug Co-Cw CEC test can be time consuming for tighter rock samples. Consequently, it is not practical for testing high clay content rocks.

The ICP CEC data is very noisy, it is obvious over exchange occurred for the carbonate samples, as well as samples containing Pyrite and k-Feldspar. An in-depth evaluation of using the ICP method for rock CEC measurement is desirable.

Nomenclature

B	Equivalent conductance of exchangeable cations	$\Omega^{-1}\cdot cm^{-1}\cdot meq^{-1}$
CEC	Cation exchange capacity	meq/100g
CEC_i	Cation exchange capacity of component i	meq/100g
Co	Total electrical conductivity of the rock–fluid system	S/m
Co-Cw	The excess conductivity due to clay-bound exchangeable cations (i.e., surface conduction)	S/m
Cw	Conductivity of (brine)	S/m
F	Formation resistivity factor (Archie)	—
F*	Modified formation resistivity factor	—
i	Index representing each component (1 to n)	—
ICP (ppm)	Concentration of cation in solution measured by ICP	ppm (mg/L)
ICP-MS	Inductively coupled plasma mass spectrometry	—
k	Permeability of the rock	mD
Q_v	Cation exchange capacity per unit pore volume	meq/cm^3
Ro	Rock Resistivity at Sw = 100%	$\Omega\cdot m$
Rw	Brine resistivity	$\Omega\cdot m$
ρ_g	Grain density of the rock	g/cm^3
φ	Porosity	fraction

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